

PAUL OSEI AFRIYIE

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EDUCATION

Bachelor of Science in Computer Science, Minor in Data Science
University of Colorado Boulder
Dean's List, Fall 2025

Expected May 2029
GPA: 3.51

TECHNICAL SKILLS

Languages	Python, C/C++, Java, JavaScript, HTML/CSS
Frameworks & Tools	Flutter, Git, Firebase, VS Code, Android Studio, Pandas, Matplotlib, PyTorch, Conda, Model Context Protocol
Data & Systems	SQL, Excel, Tableau, Linux, GitHub, Amazon Redshift, Amazon Web Services, Distributed Systems, Infrastructure as Code, Technical Writing
Organizations	ColorStack, National Society of Black Engineers

EXPERIENCE

Correll Lab, CU Boulder <i>Undergraduate Research Assistant</i>	January 2026 – Present
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- Processed 130 robotic manipulation demonstrations across 30 task/object combinations into RLDS and LeRobot formats to support Vision-Language-Action (VLA) model training.
- Visualized UR5 robot datasets using Rerun to validate quality and debug formatting issues prior to training.
- Fine-tuned SmolVLA over 15,000 training steps using bfloat16 mixed precision, reducing training loss from 0.43 to 0.02 and converging by step 10,000.

Amazon Web Services <i>Software Development Engineer Intern</i>	May 2026 – August 2026
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- Architected a multi-region, serverless asynchronous query API (API Gateway, Lambda, DynamoDB, S3), empowering 4+ internal teams to query Amazon Redshift metadata securely; engineered a custom throttling system using atomic transactions to protect production workloads.
- Built a TypeScript MCP server on top of the query API, enabling AI agents to fetch Amazon Redshift metadata via natural language while handling SigV4 authentication, async polling, and error propagation.
- Reduced Lambda cold-start initialization from ~90s to ~10s by optimizing discovery queries and externalizing schema cache to DynamoDB.
- Authored 3 technical design documents covering API architecture, throttling, and AI agent integration.

Aurora Public Schools Accountability & Research <i>Data Analyst Intern</i>	August 2024 – December 2024
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- Manipulated, cleaned, and visualized various datasets using Python, SQL, Excel, and Tableau, sourced from platforms such as the Colorado Department of Education and Infinite Campus.
- Built a Tableau dashboard analyzing an APS student dataset spanning around 59 schools, containing information such as SAT scores and attendance rates. [\[Dashboard Link\]](#)

PROJECTS

LinkedIn Gather <i>2nd Place, LinkedIn Possibilities in Tech Hackathon</i>	June 2026 [Repo Link]
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- Built a reimagined events platform in a 48-hour hackathon with a 5-person team, featuring a personalized event feed, RSVP calendar, and smart networking filters that surface mutual attendee overlap.
- Integrated the Google Gemini API to power AI-suggested talking points for "nudge" chats, helping Gen Z professionals build genuine relationships through shared real-life event participation.

CERTIFICATIONS

Google IT Automation with Python Specialization , Coursera	[Credential Link]	December 2022
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HONORS & AWARDS

Scholarships: Amazon Future Engineer, LinkedIn Possibilities in Tech, BOLD Center (CU Boulder) April 2025
President's Volunteer Service Award (Bronze), issued by President of the United States December 2024